UNDERSTAND YOUR LAND. EXPLORE YOUR FUTURE.

Team six learning farm

8.3114°, 80.4037° · 12 hectares · loam soil

Educational scenario screening. Active weather mode: **cached** (gridded estimate). Results are not validated agricultural recommendations, yield forecasts, or profit estimates.

01 / The question & the farm

Explore four-season crop rotations given 20 mm irrigation per season, 120 mm plant-available soil storage and a 120-day growing window. Planting begins in month 10, with subsequent slots six months apart.

Farmer priorities: water adequacy 26%, diversity 58%, temperature fit 16%. Previous crop: Rice. Equipment: paddy.

Local economic context: Confirm buyers for pulses before changing the rotation.

02 / Environmental evidence & scenario

NASA POWER / MERRA-2; 2023-01-01 to 2025-12-31. Daily LST. Spatial support: Meteorology 0.5° × 0.625°; solar ~1°. Null parameter values: 0.

Replay: typical; temperature shift 0°C; rainfall multiplier 1. These perturbations are hypothetical, not forecasts.

Full NASA POWER request ↗

Normalized observation SHA-256: 5e52230f8554011389af033a655a7b4c8e95e234d6c8acdb5e93e75dff69c361

03 / Compare the alternatives

Strategy	Four-season sequence	Index /100	Unmet ET demand	Constraints
Existing rotation	Rice → Rice → Maize → Rice	50.9	424 mm	Pass
Priority-led alternative · selected	Mung bean → Maize → Groundnut → Sorghum	98.6	69 mm	Pass
Water-lean alternative	Mung bean → Mung bean → Mung bean → Mung bean	56.5	0 mm	Pass

1,295 feasible generated alternatives explored. Weighted score = water adequacy × weight + diversity × weight + temperature fit × weight. This index does not establish a best real-world farm strategy.

04 / Strategy to investigate: Priority-led alternative

Mung bean → Maize → Groundnut → Sorghum

Water adequacy 94.6/100; diversity 100/100; temperature fit 100/100. 2 legume seasons and 4 family changes including the previous crop offer hypotheses to discuss, not measured soil-health improvements.

Worst fixed stress-test index: 95.2. Across four illustrative perturbations: Replay 98.6; Dry -30% 95.7; Warm +2°C 98.2; Combined 95.2.

- Legume presence indicates potential nitrogen fixation; nodulation, residues and soil tests determine actual benefit.
- Season 2: 2 replay days exceed illustrative heat threshold.

05 / Assumptions & limitations

1. Month-day weather is averaged across selected years, or replayed from the driest/wettest complete selected year. Dry/wet selection uses annual rainfall, not seasonal extremes.
2. Four consecutive half-year planting slots; fixed crop durations and constant illustrative Kc. Independent seasons reset storage to 50% of user capacity.
3. Hargreaves ET_0 is uncalibrated; Ra is converted to equivalent water depth. Rice seepage and ponding are not modeled.
4. Daily bucket: rain + uniformly distributed irrigation; overflow is discarded; no capillary rise. Soil storage is farmer-entered, not SMAP.
5. Water score is fraction of modeled ET demand supplied. Diversity combines crop variety and family breaks. Suitability counts days inside illustrative temperature limits.
6. Weights express farmer preferences; scores are indices, not probabilities or predicted effectiveness. Market notes and root traits are contextual, not scored.
7. Stress tests are hypothetical perturbations of the chosen replay; no climate forecasts or statistical confidence intervals.

06 / Questions for a local expert

- Are these crop durations, cultivars and planting dates locally feasible?
- Can soil tests and local weather records calibrate storage and crop water demand?
- Which pests or diseases are shared by the selected crops?
- Are irrigation timing, paddy water losses, drainage and erosion risks acceptable?
- Are labor, equipment, seed, buyers and financial risks manageable?

No claims are made about precise nutrient gains, future yields, profitability, soil-carbon accumulation or long-term erosion reduction. Real-world outcomes require local agronomic review and field validation.

Independent student learning reference. No NASA endorsement. Reproducible software is a starting point for scientific validation.